

03-25-2025

Mech-Design w/o Transfers:

↳ Single Object Allocation:

→ one-sided matching

↳ set A of agents

↳ set of objects $\mathbb{H}$ (cursive O in notes)

→ A matching $\mu: A \rightarrow \mathbb{H} \cup \emptyset$

↳ Assume: $|\mu(x)| = 1, |\mu^{-1}(y)| \leq 1, x \in A, y \in \mathbb{H}$

→ Mechanism:

↳ will take prefs and return a matching

↳ issue: GS thm

↳ solution: agents only care about object they get

↳ type lies in preference space w/ assigned property

→ n agents, m jobs

↳ agents care only about their job not others

Fixed Priority Mechanism (Serial Dictatorship):

- $\pi : \{1, \dots, |A|\} \rightarrow A$
- $\pi(k)$ is agent w/ k^{th} priority (assigned arbitrarily)
- Let agent choose object in order of priority
 - ↳ indirect mechanism
 - ↳ Pick best of available when your turn

Random Serial Dictatorship:

- priority ordering drawn randomly

ex:

A: $1 \succ 2 \succ \emptyset$

$$\pi(A)=1 \Rightarrow u(A)=1$$

S: $1 \succ 2 \succ \emptyset$

$$\pi(S)=2 \Rightarrow u(S)=2$$

RSD:

express $\pi(k)$ as probability of ordering

↳ lottery over outcome

Properties:

↳ no one strategic

↳ if $\exists$ indifference $\nexists$ efficiency (Pareto)

i.e. A: $1 \sim 2 \succ \emptyset$

S: $1 \succ 2 \succ \emptyset$

↳ strict preference ordering

Assume agents have strict prefs:

SD: [(My try) right path lol]

Because $\exists$ strict preferences, agent a_i
 $\pi(i)$ selects $\theta_i \in \Theta$ s.t. $\theta_i \succ \theta_k$. (all agents)

Because that is gone, $\pi(z)$ selects θ_k .

This continues $\forall a_i \in A$. This is Pareto
as reallocating necessarily makes a_i worse
off.

Proof by induction:

base: show it's true for $n=1$. assume it's true for n .
show it's true for $n+1$.

$n=1$: no way to be better off.

$n=k$: Pareto (assume)

$n=k+1$: if added, to be better off would
necessitate "stealing" from agent
ahead in priority.

With strict preferences: SD/RSD has nice properties

↳ Pareto Efficient

↳ IR: don't participate if desired

↳ "fairness" (random)

→ Read 8.1.2 in notes

↳ understand algorithm

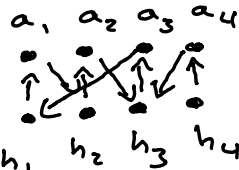
→ Agent a_i has a property right over h_i
↳ agents have strict preferences over housing

↳ top-trading cycles (8.1.2)

→ draw agents as nodes

→ "houses" " " " "

→ ask agents to point to preferred house

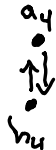


→ $a_4 \rightarrow h_3 \rightarrow a_3 \rightarrow h_1 \rightarrow a_1 \rightarrow h_2 \rightarrow a_2 \rightarrow h_3 \rightarrow a_3$

↳ no cycle

→ $a_1 \rightarrow h_2 \rightarrow a_2 \rightarrow h_3 \rightarrow a_3 \rightarrow h_1 \rightarrow a_1$

↳ cycle $\Rightarrow$ remove from chain s.t.



↳ pick preferred until pointing at self

→ there always is a cycle

→ partake in one cycle at a time

↳ no indifference